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On the Miscibility and Immiscibility of Ionic Liquids and Water

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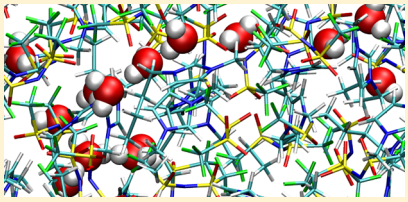
Supporting Information

ABSTRACT: Although the "like-dissolves-like" rule is often invoked to explain why sodium chloride dissolves in water, hidden behind this explanation is the delicate balance between the very large cohesive energy of the ionic crystal and large solvation energies of the ions. Room-temperature ionic liquids (ILs) are liquid analogues of ionic crystals and, as dictated by a similar energetic balance, may either fully mix with water or be immiscible with water depending on ion type and cation/anion combination. In this work, we study three hydrophobic and three hydrophilic ILs to examine whether a priori prediction of water miscibility is possible based on analysis of bulk properties alone. We find that hydrophilic and hydrophobic ILs exhibit distinct signatures in their (reciprocal space) Coulomb interactions that indicate predisposition to water mixing. Hydrophilic ILs exhibit a prominent peak in their electrostatic interactions at ∼5−8 Å length scale, largely due to repulsion between neighboring anion shells. When mixed with water, this peak is significantly reduced in magnitude, indicating that electrostatic screening by water molecules is an important driving force for mixing. In contrast, hydrophobic ILs show no such peak, indicating no predisposition to mixing. In addition to this analysis, we compute and compare solvation free energies of the six different anions in water, ion-pairing free energies at "infinitely" dilute concentration, and water absorption free energies in the different ILs. Analyzed within the context of empirical data, our calculations suggest that hydrophobicity trends of different ILs are very sensitive to precise water content at dilute conditions. For example, we predict that bis(fluorosulfonyl)imide-based ILs exhibit anomalously large water absorption free energies at zero water content, with increasing hydrophobicity as preferential absorption sites within the IL become saturated.



1. INTRODUCTION

Rationalizing the miscibility (and immiscibility) of room-temperature ionic liquids (ILs)[1−8] with common solvents is no simple task. Empirical miscibility trends are often contrary to chemical intuition; for example, many ILs are surprisingly miscible with low dielectric solvents such as chloroform, ethyl acetate, and 1,2-dichloroethane. Also puzzling are the miscibility trends of IL/water mixtures. Many ILs are hydrophilic and fully mix with water, while many other ILs are hydrophobic and phase-separate when mixed with water. While this varying hydrophilic/hydrophobic behavior of ILs is more easily rationalized for specific ion constituents [e.g., the hydrophobic, bistriflimide (TFSI) anion, or long-chain alkyl-imidazolium cations], the differing phase behavior is hard to rationalize for certain ion combinations; for example, the seemingly similar anions tetrafluoroborate (BF_4^-) and hexafluorophosphate (PF_6^-) form hydrophilic and hydrophobic ILs, respectively, when paired with 1-butyl-3-methyl-imidazolium (BMIM) cations. Additionally, the triflate ($CF_3SO_3^-$) anion forms a hydrophilic IL when paired with BMIM, despite its hydrophobic $-CF_3$ functional group. Comprehensive understanding of this IL/water mixing behavior requires detailed knowledge of the balance of complex energetic and entropic contributions, including the large solvation energies of ionic species, the hydrophobic effect associated with solvating bulky molecular ions in water, and the alteration of the neat IL electrostatic interactions due to water lubrication. In addition to fundamental insight, better understanding of IL/water miscibility is important for many separation applications in industry.[9,10]

Since the pioneering work of Seddon[10−13] and others,[9,14−16] it is now understood that the anion has the dominant influence on the hydrophobicity and polarity of prototypical ILs. Paired with relatively short-chain cations such as 1-ethyl-3-methyl-imidazolium (EMIM) or BMIM, the halides, nitrate, and acetate anions form ILs that fully mix with water, while PF_6 and TFSI anions form ILs that are insoluble and do not mix with water (hydrophobic). The BF_4 and triflate anions are "somewhere in between"[11] but are generally considered hydrophilic when paired with similar cations.[2] Although immiscible with water, hydrophobic ILs such as BMIM/PF_6 and BMIM/TFSI are still significantly hygroscopic and will absorb up to ∼0.25 mole fraction water, which is several percent by volume.[11,17] It is well-established that even a small amount of absorbed water will significantly affect the IL properties.[2,10] Such property modulation has been extensively characterized[15,18] and is not our direct focus; rather, we focus on the fundamental question of why certain ILs are fully miscible or immiscible with water.

Any adequate discussion of IL miscibility must consider the very large cohesive energies of the ILs themselves. Just as alkali-halide salts only dissolve in water because of the large and delicate counterbalance of ionic crystal cohesive energies

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(e.g., ∼−180 to −185 kcal/mol per ion pair for NaCl)[19,20] with the ion solvation free energies in water (e.g., ∼−100 kcal/mol for Na^+ and ∼−75 kcal/mol for Cl^-),[21−23] ILs will only mix with solvents if sufficiently compensating solvation energies counterbalance the large cohesive energy of the IL. Cohesive energies of prototypical ILs have been estimated by both experiment[24] and theory[25] to be ∼−120 ± 10 kcal/mol per ion pair depending on ion type; note that IL vaporization energies are significantly lower than these cohesive energies because of formation of ion pairs in the gas phase.[24,26,27] Because the cohesive energies of low-molecular-weight organic solvents are generally on the order of a few to 10 kcal/mol,[28] the mixing energetics of ILs with low dielectric solvents is dominated by the cohesive energy of the pure IL and the ion/solvent interactions (solvation energies).[29]

Water is, of course, a unique solvent. Not only is its cohesive energy (∼−10 kcal/mol) anomalously large for its molecular weight, but its unique hydrogen bonding structure is the origin of the "hydrophobic effect", giving rise to large free energy penalties for forming the necessary cavities required to solvate other molecules or clusters.[30] Therefore, in addition to the energetic contributions discussed in the preceding paragraph, the energetics associated with distortion of the liquid structure of water is an important hindrance to IL/water mixing. The relative importance of these various energetic contributions will obviously depend on the type of IL. Particular trends are intuitive: for example, for ILs composed of alkyl-imidazolium cations, increasing the length of the alkyl chain will disfavor mixing because of the hydrophobic effect/increasing amphiphilic nature of the cation[31,32] and will eventually lead to the formation of micelles and lyotropic liquid crystals for sufficiently long-chain cations;[33,34] in the latter regime, the cations have essentially become surfactants. ILs composed of such amphiphilic cations are not our focus, as these ILs exhibit intrinsic preference for self-organization/domain separation,[35−40] which predisposes them to self-assemble when mixed with water.[41,42] Additionally, outside the scope of the present discussion are so-called "protic ILs" (PILs),[43−45] which are formed by mixing a strong acid and base, with subsequent proton transfer resulting in ion formation. In PILs, water miscibility is driven by the additional energy contribution associated with the deprotonation/ion formation process in water. We note that full IL/water phase diagrams and/or IL water sorption have been characterized for a variety of systems.[17,31,32,46−50]

There has been a significant amount of previous experimental and computational work investigating IL/water mixtures, and we briefly review those studies that have targeted physical explanation of IL/water miscibility/immiscibility. Ion association and dissociation free energies in water may correlate with IL/water miscibility trends, as suggested by previous computer simulations.[51] Ion-pairing free energies are potentially inferred from concentration-dependent conductivity measurements,[52] but such analysis is subject to uncertainty.[53] Ionic conductivities have been measured for hydrophilic IL/water mixtures over the full concentration range[52,54−57] and for hydrophobic IL/water mixtures at dilute water concentration.[58] Aggregation and clustering of water in hydrophobic ILs (e.g., BMIM/TFSI) has been inferred from the measurement of diffusion coefficients, in which diffusion of ions and water show substantially different variation as a function of concentration.[59] Various motifs of anion/water/anion interactions at dilute water content were inferred from vibrational spectroscopy measurements, specifically red shifts of symmetric and antisymmetric vibrational stretches.[14,60] Previous computational studies based on the COSMO-RS model,[61] classical molecular dynamics (MD) simulations,[62] and Car−Parrinello MD simulations[63] have argued that solubility/miscibility of water/ILs is primarily governed by hydrogen bonding or other local interactions between water and the anions of the IL. We note that numerous MD simulation studies have investigated the structure and dynamics of both hydrophilic IL/water mixtures[64−74] and/or hydrophobic IL/water interfaces.[75−83]

Important information on the nature of water/IL interaction comes from vibrational spectroscopy,[60] with one of the earliest systematic studies being that of Cammarata et al.[14] For ILs composed of eight different anions and BMIM cation at dilute water concentrations, vibrational red shifts were analyzed as a measure of the relative interaction strength of water/anion hydrogen bonds; we recite the data for the specific ILs studied in this work. Red shifts (relative to water vapor) for the water antisymmetric stretch are PF_6, 84 cm^{-1}; BF_4, 116 cm^{-1}; TFSI, 119 cm^{-1}; triflate, 181 cm^{-1}; and nitrate, 236 cm^{-1}.[14] A common interpretation is that the magnitude of the red shift is proportional to the hydrogen bond strength.[14,60] Invoking this interpretation, the hydrogen bond strength between water and anions partially, but not entirely, correlates with the relative hydrophilicity of the ILs. Red shifts indicate that the triflate and nitrate anions form the strongest hydrogen bonds with water, and indeed, BMIM/triflate and BMIM/NO_3 are hydrophilic ILs. However, the magnitudes are disparate with hydrophobicity trends, as the ∼120 cm^{-1} red shift difference between the hydrophilic BMIM/BF_4 and BMIM/NO_3 ILs is significantly greater than the difference separating the hydrophobic BMIM/TFSI IL from hydrophilic BMIM/BF_4 and BMIM/triflate ILs.[60] In addition, the red shifts indicate that BF_4 and PF_6 are more similar than different in their interaction with water, so why is the former hydrophilic and the latter hydrophobic? It is important to note that the vibrational red shift in liquid water (relative to water vapor) is ∼400 cm^{-1}, which is significantly greater than in any of the ILs. Considering the findings that the heats of adsorption of water in certain ILs can be higher than or at least comparable with the heat of vaporization of liquid water,[16,84] it is clear that there must be other significant energetic contributions in addition to hydrogen bonding alone. We note that ultrafast, nonlinear spectroscopy studies[70,85−90] have provided a wealth of information on the dynamics of IL/water mixtures, and we refer the reader to these references for such discussion.

A clue to understanding IL/water miscibility is the empirical finding that trace water content has a much more significant effect on the properties of hydrophilic compared to hydrophobic ILs.[15] For example, ∼1% weight trace water dramatically reduces the viscosity of hydrophilic BMIM/BF_4,[11] but the viscosity of hydrophobic BMIM/PF_6 is altered to a much lesser extent with similar water content.[15] Relatedly, it was observed that the trace impurity of (small) chloride ions significantly alters the viscosity of hydrophilic ILs such as BMIM/BF_4 and BMIM/NO_3. For hydrophilic ILs, the trace impurities of either water or chloride are thus incorporated into the IL structure and modulate the electrostatic interactions, which in turn modulate the viscosity.[11] The significantly greater property modulation of hydrophilic compared to hydrophobic ILs due to trace impurities suggests intrinsic differences in internal liquid structure and electrostatic

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interactions. As we will show in this work, these empirical findings correlate with the different dielectric functions of the hydrophilic and hydrophobic ILs, which characterize the internal electrostatic interactions. Exceptions to these trends occur at IL/water interfaces, as sum frequency generation (SFG) spectroscopic measurements have shown that the surface of hydrophobic ILs is more perturbed by water than that of hydrophilic ILs;[91] this can be rationalized by relative partition coefficients.

Interestingly, relatively recent work has suggested that IL/water miscibility can be modulated by external electric fields.[92−94] Utilizing a combination of atomic force microscopy experiments and MD simulations, Sha et al.[92] demonstrated that the hydrophobic IL octylmethylimidazolium hexafluorophosphate (OMIM/PF_6) will spontaneously mix with water when an external electric field of ∼0.3 V/nm is applied. In a separate study, Nishi et al.[94] employed X-ray reflectivity measurement and showed that the interfacial structure of water and a hydrophobic IL, TOMA/TFPB, changes markedly upon external voltage modulation. Numerous studies have demonstrated that external electric fields can have a significant effect on the structure and dynamics of neat ILs,[95−99] and it is thus reasonable to assume similar effects for concentrated electrolyte mixtures. These observations suggest that long-range electrostatic forces play a significant role in determining IL phase behavior, including IL/water mixing, and this hypothesis is verified in this work. In particular, we will show that hydrophilic and hydrophobic ILs exhibit distinctly different internal electrostatic interactions, with hydrophilic ILs predisposed to water mixing because of the resulting electrostatic screening.

In this work, we explore physical driving forces for miscibility or immiscibility of ILs with water. We use molecular simulations to analyze six ILs and their mixtures (or dilute water concentrations if immiscible) by focusing on three "hydrophilic" ILs, BMIM/BF_4, BMIM/NO_3, and BMIM/triflate, and three "hydrophobic" ILs, BMIM/PF_6, BMIM/FSI (FSI = bis(fluorosulfonyl)imide), and BMIM/TFSI; the hydrophobic ILs are immiscible with water at significant water fraction.[11,100] We first calculate free energy quantities at dilute concentrations to assess correlation with hydrophobicity trends; these quantities include ion solvation and ion association free energies in water at infinite dilution, as well as water absorption free energies (excess chemical potential) in the different ILs (pure IL limit). We find that while these free energy quantities partially correlate with water/IL miscibility, there exist important exceptions, for example, water adsorption free energy is anomalously large in the hydrophobic BMIM/FSI IL. However, we show that analysis of the (reciprocal space) electrostatic interactions of the pure ILs allows for unambiguous classification into hydrophobic and hydrophilic categories based on signatures in the ∼1 $Å^{-1}$ wavevector region. We show that hydrophilic ILs are predisposed to water mixing because of unfavorable, repulsive electrostatic interactions at ∼1 $Å^{-1}$ inverse length scales, which are subsequently screened by water molecules. In contrast, hydrophobic ILs exhibit no such repulsive peak, and their electrostatic interactions change relatively little upon water absorption. We conclude that long-range electrostatic interactions rather than short-range hydrogen bonding are the predominant driving force for IL/water mixing. Importantly, this explanation is fully compatible with observed vibrational red shifts of water in ILs, as the rationale does not invoke discussion of hydrogen bond interaction strength, unlike previous interpretations.

2. METHODS

We employ several different MD simulation approaches to separately characterize concentration regimes of dilute IL in water, dilute water in IL, and IL/water mixtures. For clarity, we partition the Methods discussion into subsections that correspond to Results and Discussion Sections 3.1, 3.2, and 3.3

2.1. Simulations of Dilute IL in Water. Ion solvation free energies for the anions BF_4, PF_6, NO_3, triflate, TFSI, and FSI in water are computed utilizing MD simulations of a single anion solvated in a box of 5000 water molecules. Solvation free energies are computed using thermodynamic integration (TI)[22,101−107] which has been implemented with python interface to the OpenMM simulation package;[108,109] specific details of our approach, including employed lambda scaling, are described in the Supporting Information. Total sampling of 10 ns production simulation is used to compute free energies after equilibration.

All MD simulations in this work utilize the SAPT-FF force field for the ILs and IL/water interactions,[25,29,110−112] and the SWM4-NDP model is used to describe water/water interactions;[113] this hybrid approach has been previously shown to predict ionic conductivities of IL/water mixtures in excellent agreement with experiment over the full concentration range.[57] MD simulations are conducted in the *NPT* ensemble, at 1 bar, 300 K, with a Monte Carlo barostat used for pressure coupling and an extended Lagrangian, dual-thermostat scheme for temperature coupling and Drude oscillator polarization.[114] Langevin thermostats are utilized with friction coefficients of 1 ps^{-1} and mass of 0.4 au assigned to the oscillators. All simulations in this work utilize the OpenMM software[108,109] and are run on NVidia GTX-1080-Ti and TitanXp GPUs. The particle mesh Ewald (PME) method[115] is used for long-range electrostatics, and van der Waals (VDW) interactions are truncated at 1.4 nm accompanied with standard long-range correction.

Ion-pair association free energies are computed for cation/anion pairs in water for the BMIM cation and the six anions BF_4, PF_6, NO_3, triflate, TFSI, and FSI; additionally, association free energies are computed for anion/anion pairs in water for identical anion species. These simulations employ systems consisting of two ions solvated by 5000 water molecules. Potentials of mean force (PMF) describing the association free energy of two ions as a function of separation are computed utilizing umbrella sampling, followed by weighted histogram analysis.[116] The total sampling time per PMF is 60 ns after equilibration; further details are given in the Supporting Information.

2.2. Simulations of Dilute Water in IL. Free energies of water absorption in the six ILs BMIM/BF_4, BMIM/PF_6, BMIM/NO_3, BMIM/triflate, BMIM/TFSI, and BMIM/FSI are computed with TI utilizing systems of one water molecule solvated in a box of IL consisting of 200 ion pairs. Details are similar to those in Section 2.1, except that the *NPT* simulations for water in BMIM/NO_3 are run at 320 K to be sufficiently above the melting point of this IL. Additionally, 80 ns simulations are utilized to compute free energies, with this enhanced sampling necessary because of the high viscosity of the bulk ILs.

2.3. Simulations of IL/Water Mixtures. MD simulations are conducted for the pure ILs BMIM/BF_4, BMIM/PF_6,

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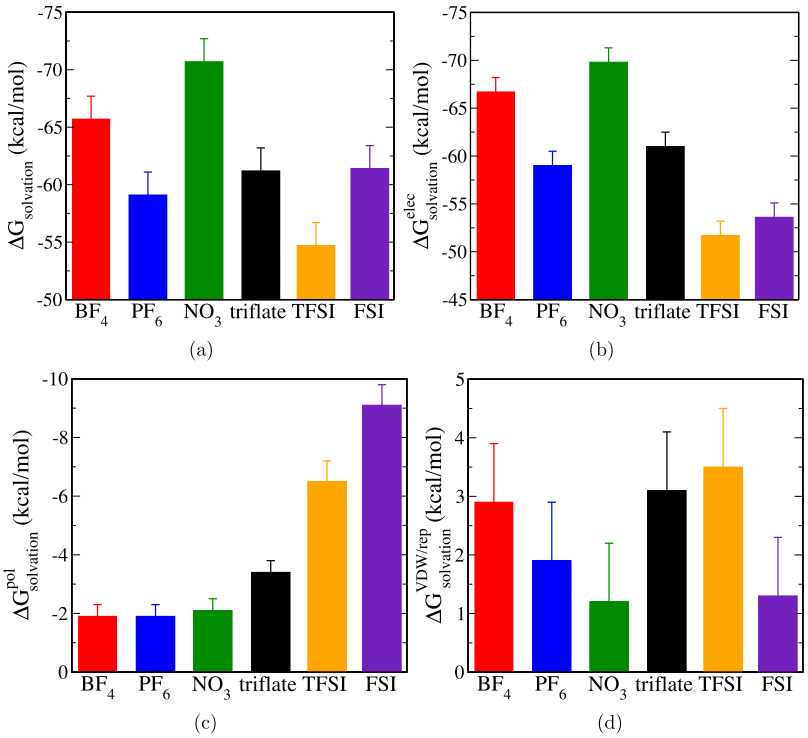

Figure 1. Solvation free energies, $\Delta G_{solvation}$, of BF_4, PF_6, NO_3, triflate, TFSI, and FSI anions in water computed using TI. Total solvation free energies, $\Delta G_{solvation}$, are shown in (a), with decomposition into contributions: (b) electrostatic, $\Delta G^{elec}_{solvation}$, (c) polarization, $\Delta G^{pol}_{solvation}$, and (d) VDW/repulsive contributions, $\Delta G^{VDW/rep}_{solvation}$.

BMIM/NO_3, BMIM/triflate, BMIM/TFSI, and BMIM/FSI and their mixtures with water to examine changes in electrostatic interactions upon water mixing. The simulation details are similar to those discussed in Sections 2.1 and 2.2, and simulations are run at 300 K, 1 bar, except for the simulations of BMIM/NO_3 and its water mixtures, which are run at 320 K (above the melting point of the pure IL). All simulations are run for 50 ns to ensure sufficient sampling of mixture heterogeneity and converged statistics for electrostatic interactions. We study IL/water mixtures of systematically increasing water content and choose systems that span IL volume fraction (ϕ_V) of $0.7 \leq \phi_V \leq 1.0$ for hydrophilic IL systems and $0.92 \leq \phi_V \leq 1.0$ for hydrophobic IL systems. These simulations utilize 220 ion pairs and appropriate number of water molecules for the specific volume fraction, ϕ_V; the exact system compositions are given in Table S1of the Supporting Information.

3. RESULTS AND DISCUSSION

We investigate whether solvation and/or ion association free energies at dilute concentrations correlate with empirical IL/water miscibility trends. In Section 3.1, we analyze solvation free energies of the BF_4, PF_6, NO_3, triflate, TFSI, and FSI anions in water, along with association free energies with the BMIM cation. In Section 3.2, we analyze water absorption free energies in the BMIM/BF_4, BMIM/PF_6, BMIM/NO_3, BMIM/triflate, BMIM/TFSI, and BMIM/FSI ILs. In Section 3.3, we investigate electrostatic interactions of the six different ILs and IL/water mixtures of systematically varying composition. Throughout this discussion, we employ the terminology "hydrophilic" for the BF_4, NO_3, and triflate anions and "hydrophobic" for the PF_6, TFSI, and FSI anions to explicitly correlate with experimental miscibility trends for the respective BMIM-based ILs.

3.1. Ion Solvation and Association Free Energies. We first discuss the solvation free energies of the BF_4, PF_6, NO_3, triflate, TFSI, and FSI anions in water. These solvation free energies, $\Delta G_{solvation}$, are expected to be large and negative as estimated by the Born equation[22] and must counterbalance the large cohesive energy of the pure IL if mixing is to occur. Computed solvation free energies, $\Delta G_{solvation}$, are shown in Figure 1a. In Figure 1b−d, we show the decomposition into electrostatic, polarization, and VDW/repulsive contributions, i.e., $\Delta G_{solvation} = \Delta G^{elec}_{solvation} + \Delta G^{pol}_{solvation} + \Delta G^{VDW/rep}_{solvation}$, with details of this decomposition given in the Supporting Information. As a benchmark, our predicted value of $\Delta G_{solvation} \approx -70 \pm 2$ kcal/mol for the nitrate anion in water may be compared to the experimental value of $\Delta G_{solvation} \approx -63$ kcal/mol,[21] the latter with typical uncertainty ∼1−2 kcal/mol.[23] We therefore consider our predictions in Figure 1 to be semiquantitative and expect that predicted trends for different ions are accurate; note that this magnitude discrepancy relative to experiment is not uncommon for solvation free energy predictions.[104]

For the six anions, the $\Delta G_{solvation}$ values span a ∼15 kcal/mol range, with the nitrate anion being the most favorably solvated ($\Delta G_{solvation} \approx -70$ kcal/mol) and the TFSI anion being the least favorably solvated ($\Delta G_{solvation} \approx -55$ kcal/mol). The BF_4 anion has the second most favorable solvation free energy ($\Delta G_{solvation} \approx -65$ kcal/mol), and the PF_6, triflate, and FSI anions are close in range $\Delta G_{solvation} \approx -59$ to -62 kcal/mol, with order PF_6 < triflate < FSI from least to most favorable solvation free energy (however, note that predicted differences between these three anions are on the order of the simulation uncertainties). The relative $\Delta G_{solvation}$ values are largely

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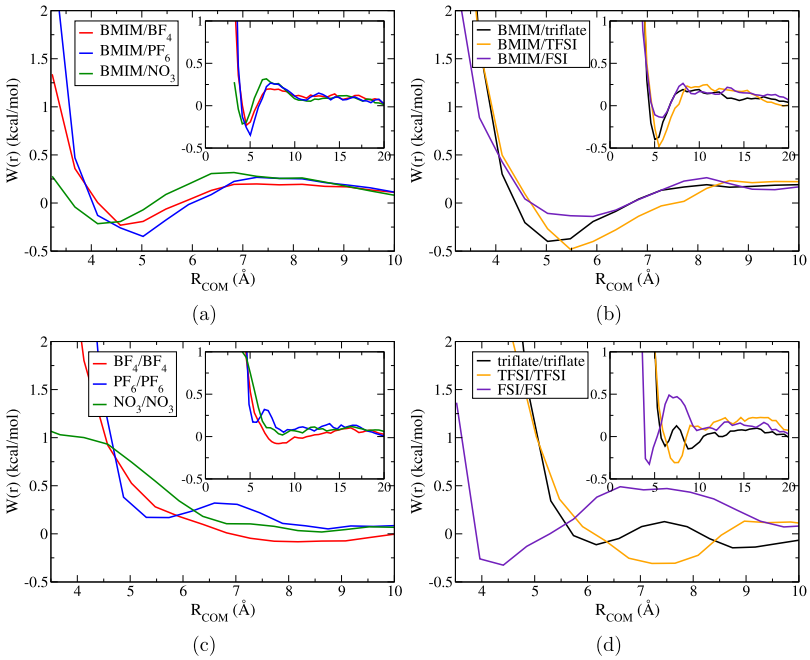

Figure 2. PMFs for cation/anion and anion/anion association in water. (a) PMFs for cation/anion association with BMIM cation and BF_4, PF_6, and NO_3 anions; (b) PMFs for cation/anion association with BMIM cation and triflate, FSI, and TFSI anions; (c) PMFs for anion/anion association for BF_4, PF_6, and NO_3 anions; (d) PMFs for anion/anion association for triflate, FSI, and TFSI anions. The graphs highlight the most interesting range of separations $\sim 3.5 \leq R_{COM} \leq 10$ Å, while the insets show the PMFs over the full computed range, $3 \leq R_{COM} \leq 20$ Å.

dictated by the electrostatic contribution, $\Delta G^{elec}_{solvation}$, with the latter given in Figure 1b. Comparing Figure 1a and b, it is seen that there is a high degree of correlation between $\Delta G_{solvation}$ and $\Delta G^{elec}_{solvation}$ values for the different ions, with the exception of the FSI anion. Trends in $\Delta G^{elec}_{solvation}$ are rationalized by the Born solvation model, which predicts that the solvation energy is inversely proportional to ion size. The only exception to this trend is PF_6 and triflate, in which triflate, although larger, is electrostatically solvated more strongly by water and is rationalized by the large dipole of the sulfonate group. The overall trend in $\Delta G^{elec}_{solvation}$ is TFSI < FSI < PF_6 < triflate < BF_4 < NO_3.

While electrostatics is the dominant contribution to $\Delta G_{solvation}$, polarization contributions become increasingly important for larger ions. Because polarization is strictly attractive, it partially counterbalances the ion size trend predicted by the Born model for the electrostatic free energy contribution. The polarization contribution, $\Delta G^{pol}_{solvation}$, is shown in Figure 1c; this term reflects the contribution of anion polarization, whereas polarization of the water molecules is implicitly present in all of the terms (see the Supporting Information). In general, the larger ions are more polarizable than the smaller ions, and this contribution can be quite significant, with $\Delta G^{pol}_{solvation} \approx -9$ kcal/mol for the FSI anion. Note that $\Delta G^{pol}_{solvation}$ results from both the polarizability of the ion and the magnitude of the local electric field from the surrounding water molecules; this local electric field will be larger in magnitude for closer approach of water molecules. Thus, $\Delta G^{pol}_{solvation}$ is slightly larger in magnitude for nitrate compared to that for PF_6, partly due to the stronger electrostatic interactions of nitrate with water, which leads to larger local electric fields because of shorter interaction distances. It is interesting to note the larger polarization contribution for FSI compared to that for TFSI, as TFSI is the larger anion. We expect that this is because the hydrophobic $-CF_3$ groups of TFSI create a "buffer" region around the ion, which reduces the number of close-contact water molecules and thus decreases the magnitude of the local electric field.

The final contribution to the ion solvation free energy is $\Delta G^{VDW/rep}_{solvation}$, which is shown in Figure 1d, and results from the VDW (dispersion) and short-range repulsive interactions. For context, it is helpful to consider the solvation free energy of methane, which is $\Delta G_{solvation} \approx 2$ kcal/mol and largely results from VDW/repulsive interactions with water (although electrostatics is not negligible[117,118]). This is interpreted as the free energy cost required to form a cavity in water,[30] balanced by the attractive dispersion interactions between methane and water. For small anions such as BF_4, PF_6, and nitrate, $\Delta G^{VDW/rep}_{solvation}$ is within ~1 kcal/mol of $\Delta G_{solvation}$ for methane. The triflate and TFSI anions exhibit slightly larger $\Delta G^{VDW/rep}_{solvation}$ values because of their larger size. Note that while the free energy cost for creating a cavity in water is expected to increase with solute size,[30] this will be partially compensated by increased VDW attraction, which is the reason for a relatively small range of free energy values for the different anions. Relative to its size, the FSI anion exhibits a rather small $\Delta G^{VDW/rep}_{solvation}$ value; however, this is consistent with the large polarization term (Figure 1c) for this anion, which in general should correlate with the magnitude of dispersion interactions. The overall better water solvation of the FSI anion compared to triflate and TFSI (Figure 1a) is thus explained by the polarization and VDW/repulsive contributions in Figure 1c,d. The hydrophobic $-CF_3$ groups of triflate and TFSI enhance the free energy penalty for cavity formation in water and reduce polarization/dispersion interactions by creating a hydrophobic "buffer" region around the ion.

The free energies of solvation of the different anions (Figure 1) indicate the relative solubility of the ILs in water at dilute concentrations. As ion concentration increases from the dilute regime, ion pairs may start to form, with the extent of ion

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pairing dependent on both concentration and the free energy of ion association. We next examine the relative ion-pairing free energies for the six different anions with BMIM cations (in the dilute regime) by computing PMFs for ion/ion association/dissociation in water. These PMFs determine the reversible work for cation/anion association and are shown in Figure 2. In Figure 2a, PMFs are shown for single ion pairs of BMIM cations with BF_4, PF_6, and NO_3 anions in water, and in Figure 2b, PMFs are shown for single ion pairs of BMIM cations with triflate, FSI, and TFSI anions in water.

All six cation/anion PMFs are characterized by relatively shallow minima of less than 1 kcal/mol and located between 4 < R_{COM} < 6 Å, where R_{COM} is the center-of-mass separation distance. These shallow well-depths (small free energy of association) may initially appear surprising, considering that gas-phase cation/anion binding energies are approximately −80 to −90 kcal/mol.[25] However, similar ion-pairing free energies have been predicted by others: for example, Raju and Balasubramanian computed PMF well-depth of 1 kcal/mol for BMIM/PF_6 ion pairs in water,[119] and Maginn and co-workers[51] reported PMF well-depth of 1 kcal/mol for BMIM/TFSI ion pairs in water; PMF well-depths for alkyl-imidazolium/halide ion pairs in water have been predicted to be shallower than 0.5 kcal/mol.[51,120] Our predictions are thus in qualitative agreement with previous results, with discrepancies rationalized by force-field differences (e.g., electronic polarization). Evidently, the shallow well-depths result from the counterbalance between large gas-phase ion interaction energies and large solvation free energies of isolated ions.

The PMF minima for different cation/anion pairs increase in attraction as FSI < NO_3 < BF_4 < PF_6 < triflate < TFSI (Figure 2a,b). These differences are rationalized by the relative strengths of cation/anion interactions and relative ion solvation and also hydrophobic effects for association of larger ions (note the aliphatic chains of the BMIM cation and $-CF_3$ groups of triflate and TFSI). The closest distance minima occur for ion pairs of BMIM and the small, planar NO_3 anion ($R_{COM} \approx 4$ Å), while the BMIM/TFSI ion pair exhibits the largest center-of-mass separation distance ($R_{COM} \approx 5.5$ Å). We attribute the shallow well-depth of the BMIM/FSI ion pair to the relatively favorable solvation of the dissociated FSI anion (Figure 1) and its weaker cation/anion interaction compared to ion pairs formed from the smaller BF_4 and nitrate anions. The primary conclusion, however, is that the PMF minima for the six different cation/anion pairs are surprisingly similar in magnitude, considering the difference in size, shape, charge delocalization, and hydrophobicity of the anions. In Figure 2, all PMFs have been normalized to zero for the largest separation distance (20 Å), so that free energy values are relative to the dissociated (solvated) ion species.

In addition to cation/anion association, it is also interesting to analyze the free energy profiles (PMFs) for anion/anion interactions in water. In the gas phase, such anion/anion interactions would be highly repulsive and uninteresting; however, water has exceptional screening ability ($\epsilon \approx 80$), and the repulsion between anions is greatly reduced in aqueous solution. In addition, one expects to see signatures of the hydrophobic effect for the larger anions, characterized by local minima in the anion/anion PMFs. In Figure 2c,d, we show computed anion/anion PMFs for the six different anions; the good screening ability of water molecules is evident, as the net repulsion between anions is generally less than 1 kcal/mol for center-of-mass separation distances greater than ∼5 Å. Qualitatively, the anion/anion interactions can be grouped into two categories based on whether they exhibit local minima in the free energy profiles. The smaller, more "hydrophilic" BF_4 and nitrate anions do not exhibit local minima in their respective anion/anion PMFs, indicating lack of any significant hydrophobic effect. The four larger anions, PF_6, triflate, FSI, and TFSI, do show signatures of the hydrophobic effect, exhibiting local minima in their corresponding anion/anion PMFs. The shape and minima of the PMFs are very similar for the PF_6 and triflate anions, with triflate anions exhibiting slightly greater tendency for association compared to PF_6 anions. The PMFs for the larger FSI and TFSI anions are distinct because of the larger size and also significant conformational flexibility of these anions. Both the FSI and TFSI anions readily interconvert between "cisoid" and "transoid" conformations,[111] which broaden the local minima in the PMFs. For the larger TFSI anion, the minima in the anion/anion PMF appear between 6 < R_{COM} < 9 Å, whereas the minima are significantly closer, 4 < R_{COM} < 6 Å, for the FSI anion/anion interaction.

Can the miscibility (or immiscibility) of the six different ILs with water be explained based on the free energy calculations discussed thus far? We propose that the answer is "partially" but not "fully". Analysis of both the anion solvation free energies (Figure 1) and ion association free energies (PMFs, Figure 2) clearly indicates that BF_4 and nitrate are the most hydrophilic anions, in agreement with the fact that BMIM/BF_4 and BMIM/NO_3 are fully miscible with water. While the four remaining anions appear more hydrophobic, the quantitative free energy differences are not large enough to a priori predict immiscibility with water, without considering differences (5−10 kcal/mol per ion pair) in the IL cohesive energies.[112] Furthermore, both the solvation free energy and ion association free energy calculations suggest that the triflate anion is similar to the three hydrophobic anions (PF_6, FSI, TFSI) in terms of its interaction with water compared to the hydrophilic BF_4 and nitrate anions. Thus, the free energy quantities in Figures 1 and 2 do not explain why BMIM/triflate mixes with water, but BMIM/PF_6 and BMIM/FSI do not. For further insight, we next examine the absorption free energies of water in the six different pure ILs.

3.2. Water Absorption Free Energies in ILs. All ILs are significantly hygroscopic, regardless of whether they are fully miscible with water. For example, the hydrophobic ILs BMIM/TFSI and BMIM/PF_6 absorb ∼0.25 mole fraction of water at saturation (room temperature), which is several percent water by volume.[11,17] It is reasonable to assume that the water absorption of ILs at low humidity would correlate with their water miscibility; indeed, Cao et al.[49,121] found the trend BMIM/PF_6 < BMIM/TFSI < BMIM/BF_4 < BMIM/triflate < BMIM/NO_3 for water uptake (by weight) at 52% relative humidity. Interestingly, however, the water content of vacuum-dried ILs does not necessarily correlate with hygroscopic trends at low relative humidity. For example, O'Mahony et al.[122] found that vacuum-dried BMIM/BF_4 contained less water than both vacuum-dried BMIM/PF_6 and BMIM/TFSI, with BMIM/PF_6 also containing more water than BMIM/triflate. Although there is inevitable uncertainty with such measurements, these data indicate that ILs may exhibit different regimes of hydrophobicity/hydrophilicity at low hydration. In other words, trends in the excess chemical potential of water for different ILs may depend sensitively on trace water content. We illustrate an example of this below, for

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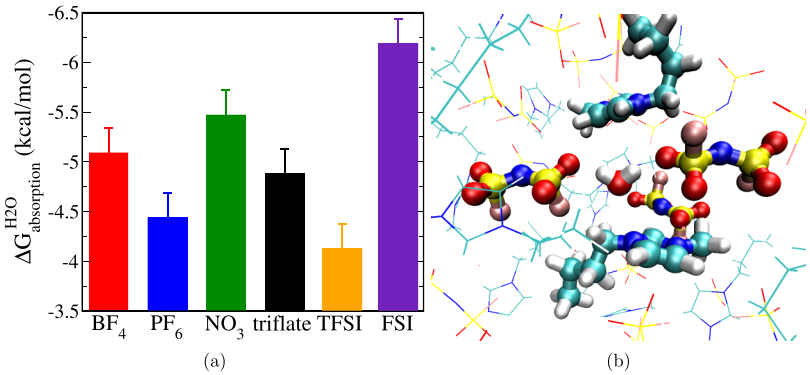

Figure 3. (a) Water absorption free energies $\Delta G^{H_2O}_{absorption}$ in BMIM/BF_4, BMIM/PF_6, BMIM/NO_3, BMIM/triflate, BMIM/TFSI, and BMIM/FSI ILs at 300 K, except for BMIM/NO_3 which is at 320 K (see Methods). Uncertainties are shown with error bars and are on the order of a couple tenths of a kcal/mol. (b) Simulation snapshot of a water molecule solvated in BMIM/FSI, with elements colored as H: white, C: cyan, N: blue, O: red, F: pink, S: yellow. Hydrogen bond distances between water hydrogen atoms and FSI oxygen atoms are ∼1.75−1.9 Å.

which we find that the excess chemical potential of water in the pure BMIM/FSI IL is anomalous relative to the hydrophobicity of this IL at higher water content.

Utilizing TI, we compute the excess chemical potential of water in the six different (pure) ILs, which we denote as $\Delta G^{H_2O}_{absorption}$. In Figure 3, we show computed $\Delta G^{H_2O}_{absorption}$ values for BMIM/BF_4, BMIM/PF_6, BMIM/NO_3, BMIM/triflate, BMIM/TFSI, and BMIM/FSI. We first compare our results to previous work and then discuss trends. $\Delta G^{H_2O}_{absorption}$ values have been determined from experimental data for BMIM/PF_6 and related 1-octyl-3-methylimidazolium (C8MIM) ILs and are −3.9 kcal/mol for BMIM/PF_6, −3.6 kcal/mol for C8MIM/PF_6, and −4.4 kcal/mol for C8MIM/BF_4 (at room temperature).[84] Using Monte Carlo simulations, Shah and Maginn predicted $\Delta G^{H_2O}_{absorption}$ values of −4.4 kcal/mol for BMIM/PF_6[123] and −4.5 kcal/mol for the related 1-hexyl-3-methylimidazolium (C6MIM)/TFSI IL.[124] Deschamps et al. predicted a larger value of $\Delta G^{H_2O}_{absorption} \approx -5.5$ kcal/mol for BMIM/PF_6.[125] In addition, Lynden-Bell and co-workers have computed a value of $\Delta G^{H_2O}_{absorption} = -6.9$ kcal/mol in the IL dimethylimidazolium chloride at 400 K.[84] Because of the larger cohesive energy and stronger water interactions with the chloride anion, it is expected that $\Delta G^{H_2O}_{absorption}$ for dimethylimidazolium chloride should be larger (in magnitude) than for the other ILs studied. For the water absorption free energy in BMIM/PF_6, we compute a value of −4.4 kcal/mol which is in remarkably close agreement with the prediction of Shah and Maginn and is fairly close to the experimental value (−3.9 kcal/mol).[84] If we assume a similar trend with cation chain length for BMIM/BF_4 as is experimentally observed for the PF_6 ILs, our prediction of −5.1 kcal/mol for the water absorption free energy in BMIM/BF_4 is in fairly good agreement with the extrapolated "experimental" value of −4.7 kcal/mol, considering both simulation and experimental uncertainties.

We find interesting trends in the predicted $\Delta G^{H_2O}_{absorption}$ values for the six ILs shown in Figure 3a. There is some degree of correlation between the $\Delta G^{H_2O}_{absorption}$ values with the hydrophobicity of the ILs; for example, hydrophilic BMIM/BF_4, BMIM/triflate, and BMIM/NO_3 exhibit larger (in magnitude) $\Delta G^{H_2O}_{absorption}$ values than does the hydrophobic IL BMIM/TFSI. However, the most notable exception is that the hydrophobic BMIM/FSI exhibits the largest (in magnitude) $\Delta G^{H_2O}_{absorption}$ value of all six ILs! Interpretation of this finding benefits from inspection of simulation snapshot of water coordination environment within BMIM/FSI, as shown in Figure 3b. In this configuration, the water molecule forms two close-contact hydrogen bonds (∼1.75−1.9 Å distance) with the oxygen atoms of the FSI sulfonyl groups, and the water oxygen atom is closely situated above the imidazolium ring of the cation. We note that strong hydrogen bonds between water and anion sulfonyl groups have been inferred from SFG spectra for similar ILs.[85] Clearly, this ionic environment is optimal for absorption/interaction with a water molecule, and we assume that the free energy of absorption will significantly drop in magnitude after all such absorption sites are "occupied" by water molecules with increasing water content.

Our computed $\Delta G^{H_2O}_{absorption}$ values for the six ILs, along with absorption site analysis depicted in Figure 3, suggest that hygroscopic trends may be highly sensitive to the precise water content of the IL; it follows that the hydrophobicity of the IL as determined by its water miscibility may not reflect the water/IL interaction at trace water content and vice versa. This interpretation is both consistent with and in fact clarifies experimental findings. As mentioned, O'Mahony et al.[122] found that trends in IL water content were very different for vacuum-dried systems compared to systems equilibrated at 52% relative humidity. Additionally, high sensitivity of hydrophobicity trends to precise water content explains why anion/water hydrogen bond strengths at dilute concentrations (as determined by IR red shifts) do not necessarily correlate with miscibility trends.[14] Hydrophobicity trends of ILs should thus be separately discussed and characterized in at least two regimes: the pure IL limit and the limit of significant water content (e.g., several to tens of percent water by volume fraction). In the next section, we show that IL/water miscibility trends fully correlate with the extent of modulation of the cohesive energy and electrostatic screening of the pure IL upon water mixing.

3.3. Modulation of IL Electrostatics upon Water Mixing. Analysis of solvation and ion association free energies in water and water solvation free energies in ILs does not quantify the modulation of IL cohesive energies upon water

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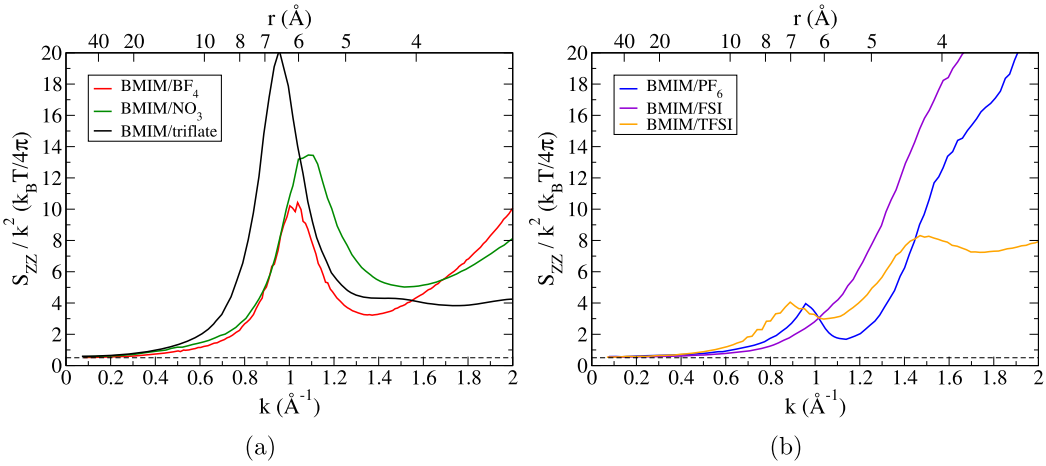

Figure 4. Electrostatic energies of (a) hydrophilic ILs and (b) hydrophobic ILs decomposed into Fourier contributions, S_{ZZ}/k^2, in units of $k_BT/4\pi$. Simulations are at 300 K, except for BMIM/NO_3 which is run at 320 K, such that it is above its melting point. The top axis depicts the corresponding real-space distance, and the asymptotic limit is indicated by the dashed line.

mixing. IL cohesive energies are very large, approx. −120 ± 10 kcal/mol,[25] because of the electrostatic interactions between ions.[126] The propensity of ILs to mix with water will thus depend on the electrostatic properties of the neat IL and the modulation of these electrostatic interactions upon water mixing. Water molecules may break up and screen interactions between ions in the IL, particularly repulsive interactions between smaller anions. Furthermore, strong ion/dipole interactions are formed between the ions and individual water molecules, which is an important driving force, and the magnitude of such interactions depends on ion type. We show that hydrophilic and hydrophobic ILs exhibit qualitatively distinct electrostatic signatures, with hydrophilic ILs predisposed to mixing with water because of inherently unfavorable anion/anion repulsion that is effectively screened by water molecules.

We first discuss the electrostatic properties of the neat ILs and then examine how these interactions are modulated by water. The electrostatic interactions within the liquid are most easily analyzed in Fourier space. For context, we note the following transformation of the electrostatic energy of the system from a real-space to Fourier space analysis

$$E_{\text{elec}} = \frac{1}{2}\int \mathrm{d}r_1\, \mathrm{d}r_2 \frac{\rho(r_1)\rho(r_2)}{r_{12}} = \frac{1}{2V}\sum_k \frac{4\pi\rho_k\rho_{-k}}{k^2} \tag{1}$$

In this expression, $\rho(r)$ is the charge density of the liquid, ρ_k is a Fourier component of this charge density, and the factor of one-half corrects for double counting interactions. We note that this expression contains self-interaction contributions, but these may be subtracted off and are unimportant for the following discussion. To simplify notation, we define the charge−correlation structure factor as

$$S_{ZZ}(k) = \frac{1}{V}\langle \rho_k\rho_{-k}\rangle \tag{2}$$

where the ⟨...⟩ denotes an ensemble average. The quantity S_{ZZ}/k^2 then reflects the average contribution to the electrostatic energy of each Fourier component (wavevector). The k-dependence of this quantity describes the relative contribution of electrostatic interactions over various length scales. In Figure 4, we plot these electrostatic interactions (for each Fourier component) for the six different ILs, which are computed similar[112] to the PME approach.[115] Electrostatic interactions for the hydrophilic ILs (BMIM/BF_4, BMIM/NO_3, BMIM/triflate) are plotted in Figure 4a and for hydrophobic ILs (BMIM/PF_6, BMIM/FSI, BMIM/TFSI) in Figure 4b. These electrostatic interactions are plotted in units of $k_BT/4\pi$, as the long wavelength limit ($k \to 0$) is set by the Stillinger−Lovett sum rule,[127,128] indicated by the dashed line in the figure. It is clear that there are distinct differences in the electrostatic interactions of the hydrophilic and hydrophobic ILs. The hydrophilic ILs exhibit a pronounced peak in the electrostatic interactions S_{ZZ}/k^2 at wavevectors $0.8 < k < 1.2$ Å^{-1}, corresponding to real-space distances of between ∼5 and 8 Å. For the hydrophobic ILs, such a peak is either not present or is significantly reduced relative to the hydrophilic ILs. This peak makes a repulsive contribution to the cohesive energy and should be interpreted as resulting from unfavorable electrostatic interactions. We note that the plotted electrostatic energies are strictly repulsive; these energies only become attractive when the self-interaction contribution is subtracted out and short-range real-space interactions are added in.[115]

What explains the $0.8 < k < 1.2$ Å^{-1} repulsive peak in the electrostatic interactions of the hydrophilic ILs and the lack of such a peak for hydrophobic ILs? In previous work,[127,128] we have shown that this peak is a result of both anion/anion repulsion and depletion of cation/anion attraction (at this length scale). Taking BMIM/BF_4 as an example, the distance between BF_4 anions in neighboring ion shells is ∼6−7 Å,[127] which directly correlates with the location of the repulsive electrostatic peak (see top, real-space axis in Figure 4). This distance regime thus reflects the molecular detail of the IL, indicating the packing of anions and cations and matching of ion size, shape, and charge distribution. For the ILs composed of smaller anions (BF_4, NO_3, triflate), the peak in electrostatic interactions indicates imbalance of counterion attraction and like-ion repulsion at these length scales, with the latter being dominant. The ILs composed of larger anions (FSI, TFSI) exhibit less net repulsion at ∼5−8 Å (lack of or reduced peak in S_{ZZ}/k^2 at $0.8 < k < 1.2$ Å^{-1}) presumably due to the more diffuse anion charge and better size match with the cation.

We suggest that the electrostatic interactions shown in Figure 4 determine the predisposition for the ILs to mix with water. The repulsive peak in S_{ZZ}/k^2 of hydrophilic ILs indicates that mixing with water would be energetically favorable if water

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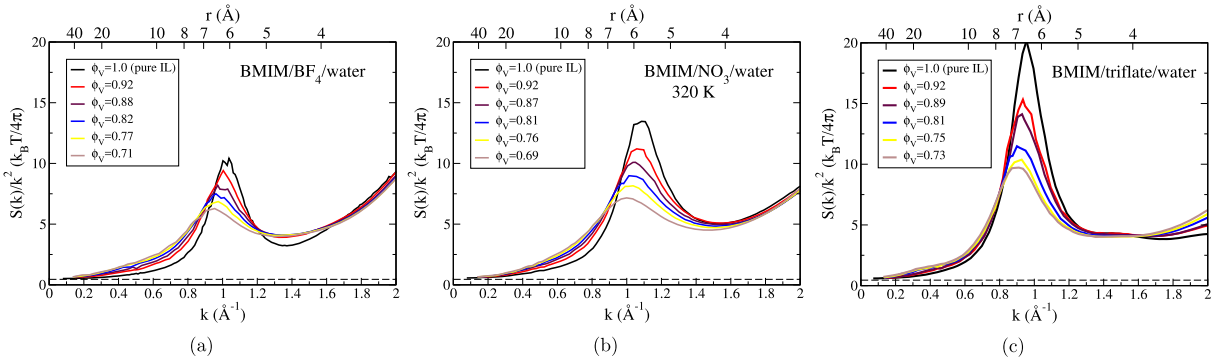

Figure 5. Electrostatic energies of IL/water mixtures for hydrophilic ILs (a) BMIM/BF_4, (b) BMIM/NO_3, and (c) BMIM/triflate as a function of IL volume fraction, ϕ_V. Electrostatic energies are decomposed into Fourier components, S_{ZZ}/k^2, in units of $k_BT/4\pi$. BMIM/NO_3/water mixtures are simulated at 320 K (Methods), while other simulations are at 300 K. The top axis depicts the corresponding real-space distance, and the asymptotic limit is indicated by the dashed line.

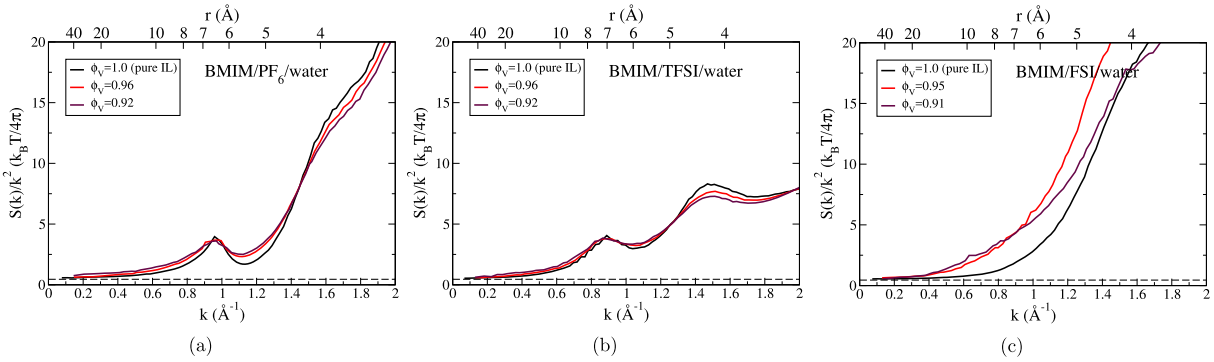

Figure 6. Electrostatic energies of IL/water mixtures for hydrophobic ILs (a) BMIM/PF_6, (b) BMIM/TFSI, and (c) BMIM/FSI, at 300 K, as a function of IL volume fraction, ϕ_V. Electrostatic energies are decomposed into Fourier components, S_{ZZ}/k^2, in units of $k_BT/4\pi$. The top axis depicts the corresponding real-space distance, and the asymptotic limit is indicated by the dashed line.

molecules were to screen these repulsive (anion/anion) interactions. In contrast, the hydrophobic ILs have no such peak, indicating no systematically unfavorable interactions to screen. Thus, we expect that when mixed with water, the electrostatic peak in S_{ZZ}/k^2 of hydrophilic ILs will decrease in magnitude because of water screening, providing an energetic driving force for mixing. To test this hypothesis, we perform the same electrostatic analysis on IL/water mixtures of systematically varying water content. Because the ILs are composed of different sized ions, we classify the mixtures by volume fraction (rather than mole fraction[57]) of IL by focusing on systems with $\sim 0.7 < \phi_V < 1.0$, where $\phi_V = 1.0$ indicates pure IL.

In Figure 5, we show electrostatic interactions, S_{ZZ}/k^2, for IL/water mixtures composed of the hydrophilic ILs: analysis of BMIM/BF_4/water mixtures is shown in Figure 5a, BMIM/NO_3/water mixtures in Figure 5b, and BMIM/triflate/water mixtures in Figure 5c. For all of these mixtures, we find that the repulsive electrostatic peak at $0.8 < k < 1.2$ Å^{-1} is significantly reduced as more water is added to the mixture. This effect is evident even at low water content, with the $\phi_V \approx 0.9$ fraction mixtures exhibiting noticeably modulated electrostatics. The interpretation is that water molecules screen the unfavorable repulsive interactions in the IL, which intrinsically originated from the small size and localized charge of the anions. For the low water content mixtures, water acts as a lubricant, structurally mixing with the cations and anions to screen repulsive interactions and reduce the net electrostatic repulsion at the $0.8 < k < 1.2$ Å^{-1} distance regime. We conclude that this is a primary driving force for mixing between water and the hydrophilic ILs. Note that this driving force is energetic (electrostatic) and results from the intrinsic electrostatic properties of the neat IL. The hydrophilic ILs are thus predisposed to mix with water because of the repulsive peak at $0.8 < k < 1.2$ Å^{-1} in their reciprocal space electrostatic interactions.

The hydrophobic ILs exhibit no significant peak in their electrostatic interactions at $0.8 < k < 1.2$ Å^{-1} length scale. It is thus interesting to analyze how the electrostatic interactions in low water content systems of these ILs are affected (or unaffected) by the presence of water compared to the hydrophilic ILs. Although immiscible with water, hydrophobic ILs are still highly hygroscopic, absorbing significant water before separating into distinct phases; for example, BMIM/TFSI and BMIM/PF_6 saturate at about ~0.25 mol fraction of water (~3% volume fraction) at room temperature.[11,17] We thus restrict our analysis to mixtures of less than 10% water by volume, as this upper bound corresponds to highly supersaturated solutions for which water domain formation is apparent, predecessing phase separation. Electrostatic interactions in these hydrophobic IL/water "mixtures" are shown in Figure 6a for BMIM/PF_6/water, Figure 6b for BMIM/TFSI/water, and Figure 6c for BMIM/FSI water mixtures. Unlike the hydrophilic ILs, in which the $0.8 < k < 1.2$ Å^{-1} electrostatic peak was significantly modulated by the presence of water, there is little change in these electrostatic interactions for hydrophobic BMIM/PF_6 and BMIM/TFSI at low water content. The lack of modulation is expected based on analysis

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of the bulk hydrophobic ILs, as the absence of a repulsive electrostatic peak suggests no systematic need for water screening. The hydrophobic BMIM/FSI IL exhibits a subtle but interesting shift in its electrostatic interactions in the presence of water (Figure 6c), which is due to screening and is similar to analogous shifts resulting from electronic polarization.[112] However, this water modulation in BMIM/FSI is very different and more subtle than the modulation in the hydrophilic ILs. On the basis of this analysis, we conclude that the hydrophobic ILs are not predisposed to mix with water, as electrostatic interactions exhibit no obvious energetic relaxation upon water absorption and screening (unlike the hydrophilic ILs). The large cohesive energy of the IL and free energy cost of water cavity formation prevent the hydrophobic ILs and water from mixing in the absence of the clear energetic driving force that was observed for hydrophilic IL/water mixing.

4. CONCLUSIONS

We have shown that free energetic quantities for dilute IL/water systems do not always correlate with the experimental IL/water miscibility behavior, but rather the relative "hydrophobicity" of different ILs may be quite sensitive to the precise water content. For example, the "hydrophilic" triflate and "hydrophobic" PF_6 anions exhibit comparable solvation free energies, cation/anion association free energies (with BMIM cations), and anion/anion association free energy profiles in water at dilute ion content. None of these metrics allow a priori prediction of the empirical fact that BMIM/triflate is fully miscible with water, whereas BMIM/PF_6 is not. Furthermore, the "hydrophobic" anion FSI appears more "hydrophilic" at the concentration limits investigated in Sections 3.1 and 3.2, than would be suggested by the empirical fact that BMIM/FSI is immiscible with water. For example, we predict that the FSI anion has a larger magnitude solvation free energy in water than does the triflate anion (with the caveat that differences are close to statistical error). Most surprising is our prediction of the absorption free energy of water in pure BMIM/FSI, which is more favorable than that in any other IL studied, including the hydrophilic ILs.

Numerous empirical observations suggest that hydrophobicity trends of ILs depend sensitively on trace water content. Notably, trends in water absorption are different for vacuum-dried ILs compared to that for ILs in humid air,[122] and IR measurements of hydrogen bond strength between anions and water under dilute concentrations do not entirely correlate with IL miscibility trends.[14] Our predictions for the absorption free energy of water in different ILs suggest a microscopic interpretation of these empirical data. As shown for BMIM/FSI, ILs exhibit a heterogeneous distribution of coordination sites for absorbed water; at low water content, water is absorbed in the most strongly coordinating environment within the liquid. However, ion coordination becomes more restricted with increasing water content and may exhibit significantly different energetics than predicted for the pure IL regime. Thus, BMIM/FSI is predicted to be relatively hydrophilic at zero water content, while its hydrophobicity increases rapidly with trace hydration as the strongly preferential ionic coordination sites become saturated.

What then is the most distinguishing feature of hydrophobic and hydrophilic ILs that correlates with their empirical miscibility trends? We conclude that long-range electrostatic interactions rather than short-range hydrogen bonding are the predominant driving force for IL/water mixing, as the most distinguishing feature of hydrophilic/hydrophobic ILs is the presence/absence of a repulsive peak in the IL electrostatic interactions at $0.8 < k < 1.2$ Å^{-1} length scales. Hydrophilic ILs are predisposed to mix with water based on their bulk electrostatic properties. In hydrophilic ILs, anion/anion repulsive interactions at ∼5−8 Å distances are not fully screened by cation/anion attractions, resulting in the prominent electrostatic peak of S_{ZZ}/k^2. We have shown that upon mixing with water, water lubricates the IL and screens these repulsive interactions, reducing the magnitude of the repulsive electrostatic peak. Electrostatic analysis of the *bulk ILs* thus allows for a priori *prediction* of whether a particular IL will mix with water. Hydrophobic ILs do not exhibit analogous peaks in their reciprocal space electrostatic interactions, indicating no predisposition for water screening. Without this compensating energetic relaxation upon mixing, hydrophobic ILs are immiscible with water because of their large cohesive energies combined with the free energy cost of requisite cavity formation in water.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.9b02187.

System compositions of IL/water mixtures discussed in Sections 2.3 and 3.3; discussion of the TI method, implementation, and specific considerations for the SAPT-FF force field; benchmarks of TI implementation for chloride in water using the OPLS-AA force field and methane in water using the SAPT-FF force field; and details of PMF calculations (PDF)

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Notes

The authors declare no competing financial interest.

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